

Information Security (CS3002)

Date: November 7, 2025

Course Instructor (s)

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Sessional-II Exam

Total Time: 1 Hours

Total Marks: 15

Total Questions: 02

Semester: FALL-2025

Campus: Karachi

Dept: CS/SE/CY

Student Name

Roll No

Section

Student Signature

Instructions:

- Each question is set to have a specific answer utmost in 1-2 sentences. Unrelated longer answers will not be read nor graded nor responded to during review.

CLO # 4: Identify appropriate techniques to tackle and solve problems of real life in the discipline of information security.

Q1. [8 Marks]

- a. Consider the following SQL query intended to verify user login credentials:

SELECT *

FROM Users

WHERE username = 'Susername' AND pin = 'Spin';

- i. **Explain** how an attacker could supply crafted input values to perform a successful SQL injection attack that bypasses the authentication check.
Describe the specific input the attacker might use and explain why this causes the query to succeed without a valid password. (2 Marks)
- ii. The organization stores employee data in an encrypted database and needs to support range queries such as: **SELECT * FROM Employee WHERE eid < 300;**
When using indexes on encrypted data, the order of values can sometimes be revealed. **Explain** what order leakage is, and **describe** how index randomization helps hide this order information while still allowing range queries to work. (2 Marks)
- b. Employees at TechSolve Ltd. notice their systems slowing down and strange network activity. Later, company files become inaccessible due to encryption. Investigators find that the infection began when one employee installed a free screen-recording app. The malware spread to other computers automatically, hid itself from antivirus tools, and allowed remote attackers to use the systems for sending spam.
- i. **Identify and explain** three different types of malware shown in this scenario. (2 Marks)
- ii. In the TechSolve Ltd. attack, the malware hid from antivirus tools. **What** is a rootkit, and **which** two modes of operation could it use to stay hidden? (2 Marks)

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CLO # 3: Analyze and apply various security and risk management tools for achieving information security and privacy.

Q2. [7 Marks]

- a. A hospital database contains tables for patients, doctors, nurses, and billing staff. Access control must ensure that doctors can view and update only their assigned patients' data, while billing staff handle financial records only.
 - i. **Explain** how Capability lists and Access Control Lists (ACLs) can be implemented in this system to define and enforce permissions for doctors, nurses, and billing staff. **(2 Marks)**
 - ii. In Attribute-Based Access Control (ABAC), identify and briefly **describe** the three categories of attributes that are used to make access control decisions. **(2 Marks)**
- b. A company uses a remote challenge–response authentication system where passwords are stored as SHA-256 (salt || password).
 - i. **Outline** the key steps of the remote challenge–response authentication process between a user and a server. **(2 Marks)**
 - ii. **Explain** how adding salt to stored password hashes helps protect against rainbow table and precomputed hash attacks. **(1 Marks)**